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Université: Institut Universitaire des Sciences (IUS)

TD N° 5: Systèmes 1

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Niveau: 3^{ème} Année

Date: Le 14/Jan/2026

Travaux Dirigés – Administration Système Linux

Mise en place et sécurisation d'un serveur Samba

Entreprise: TechCorp

1. Contexte professionnel

L'entreprise **TechCorp**, spécialisée dans le développement logiciel, ouvre un nouveau bureau composé de trois équipes:

- **DEV** : Développeurs
- **SEC** : Sécurité informatique
- **OPS** : Administration systèmes et réseaux

Chaque équipe est composée de 5 utilisateurs, soit 15 utilisateurs au total.

En tant qu'administrateur système junior, ma mission consiste à concevoir, déployer et sécuriser une infrastructure serveur Linux, permettant le partage de fichiers entre les équipes via Samba.

2. Objectifs du projet

- Mettre en place un serveur Samba
- Créer des partages publics et privés
- Gérer les droits d'accès par groupe
- Permettre la collaboration interne
- Mettre en place un dossier de transfert inter-équipes
- Tester l'accès depuis un client Windows
- Réaliser un audit du serveur Samba

3. Mise en place du serveur Samba

3.1 Installation du service Samba

`sudo apt update`

`sudo apt install samba`

`sudo apt install nfs-kernel-server`

Ubuntu-Evenort [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jan 12 11:48

marco@Ubuntu-Evenort: ~

```
Created symlink /etc/systemd/system/samba.service → /usr/lib/systemd/system/samba-ad-dc.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/samba-ad-dc.service → /usr/lib/systemd/system/samba-ad-dc.service.  
Processing triggers for ufw (0.36.2-6) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
marco@Ubuntu-Evenort:~$ sudo apt install nfs-kernel-server  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following package was automatically installed and is no longer required:  
  libllvm19  
Use 'sudo apt autoremove' to remove it.  
The following additional packages will be installed:  
  keyutils libevent-core-2.1-7t64 nfs-common rpcbind  
Suggested packages:  
  open-iscsi watchdog  
The following NEW packages will be installed:  
  keyutils libevent-core-2.1-7t64 nfs-common nfs-kernel-server rpcbind  
0 upgraded, 5 newly installed, 0 to remove and 182 not upgraded.  
Need to get 612 kB of archives.  
After this operation, 2,103 kB of additional disk space will be used.  
Do you want to continue? [Y/n]
```

11:48 AM
1/12/2026

Ubuntu-Evenort [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jan 12 11:49

marco@Ubuntu-Evenort: ~

```
182 packages can be upgraded. Run 'apt list --upgradable' to see them.  
marco@Ubuntu-Evenort:~$ sudo apt install samba  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following package was automatically installed and is no longer required:  
  libllvm19  
Use 'sudo apt autoremove' to remove it.  
The following additional packages will be installed:  
  attr libboost-iostreams1.83.0 libboost-locale1.83.0 libboost-thread1.83.0  
  libcephfs2 librados2 librdmacm1t64 liburing2 python3-dnspython python3-gpg  
  python3-ldb python3-markdown python3-samba python3-talloc python3-tdb  
  samba-ad-provision samba-common samba-common-bin samba-dsdb-modules  
  samba-vfs-modules tdb-tools  
Suggested packages:  
  python3-trio python3-aiouic python3-h2 python3-httpx python3-httpcore  
  python-markdown-doc bind9 bind9utils ctdb ldb-tools ntp | chrony winbind  
  heimdal-clients  
The following NEW packages will be installed:  
  attr libboost-iostreams1.83.0 libboost-locale1.83.0 libboost-thread1.83.0  
  libcephfs2 librados2 librdmacm1t64 liburing2 python3-dnspython python3-gpg  
  python3-ldb python3-markdown python3-samba python3-talloc python3-tdb samba  
  samba-ad-provision samba-common samba-common-bin samba-dsdb-modules  
  samba-vfs-modules tdb-tools
```

11:49 AM
1/12/2026

```
marco@Ubuntu-Evenort:~$ pwd
/home/marco
marco@Ubuntu-Evenort:~$ sudo apt install nfs-kernel-server
[sudo] password for marco:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nfs-kernel-server is already the newest version (1:2.6.4-3ubuntu5.1).
The following package was automatically installed and is no longer required:
  liblvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 182 not upgraded.
marco@Ubuntu-Evenort:~$
```

```
rpc-gssd.service is a disabled or a static unit, not starting it.
rpc-statd.service is a disabled or a static unit, not starting it.
rpc-statd.service is a disabled or a static unit, not starting it.
rpc-svcgssd.service is a disabled or a static unit, not starting it.
Setting up nfs-kernel-server (1:2.6.4-3ubuntu5.1) ...
Created symlink /etc/systemd/system/mountd.service.requires/fsidd.service →
/usr/lib/systemd/system/fsidd.service.
Created symlink /etc/systemd/system/nfs-server.service.requires/fsidd.service →
/usr/lib/systemd/system/fsidd.service.
Created symlink /etc/systemd/system/nfs-client.target.wants/nfs-blkmap.service →
/usr/lib/systemd/system/nfs-blkmap.service.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service →
/usr/lib/systemd/system/nfs-server.service.
nfs-mountd.service is a disabled or a static unit, not starting it.
nfsdclld.service is a disabled or a static unit, not starting it.
Creating config file /etc/exports with new version
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
marco@Ubuntu-Evenort:~$ samba --version
Version 4.19.5-Ubuntu
marco@Ubuntu-Evenort:~$
```

- Ces images illustre l'installation du serveur Samba (4 à 5).

4. Gestion des utilisateurs et groupes

4.1 Création des groupes

```
sudo groupadd dev
```

```
sudo groupadd sec
```

```
sudo groupadd ops
```

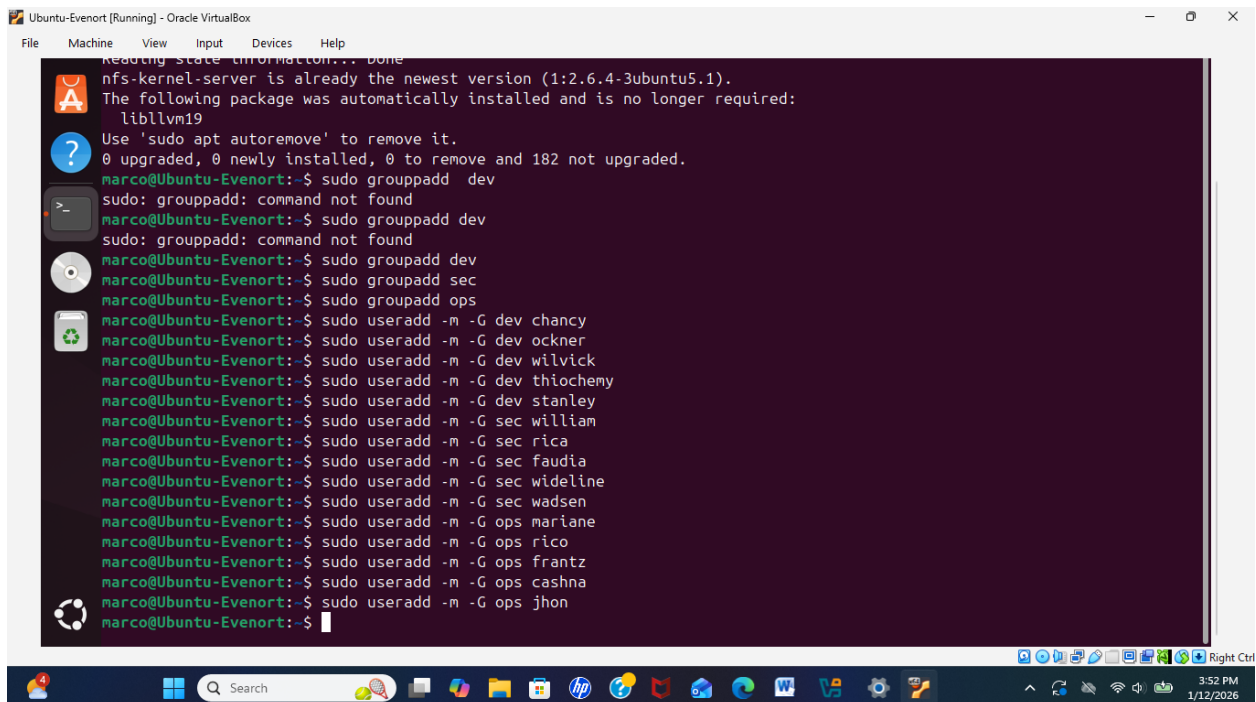
4.2 Création des utilisateurs

Chaque utilisateur est affecté à son groupe respectif.

```
sudo useradd -m -G dev dev1
```

```
sudo useradd -m -G sec sec1
```

```
sudo useradd -m -G ops ops1
```



The screenshot shows a terminal window titled 'Ubuntu-Evenort [Running] - Oracle VM VirtualBox'. The terminal output displays the following commands and their results:

```
marco@Ubuntu-Evenort:~$ sudo groupadd dev
marco@Ubuntu-Evenort:~$ sudo groupadd dev
marco@Ubuntu-Evenort:~$ sudo groupadd dev
marco@Ubuntu-Evenort:~$ sudo groupadd sec
marco@Ubuntu-Evenort:~$ sudo groupadd ops
marco@Ubuntu-Evenort:~$ sudo useradd -m -G dev chancy
marco@Ubuntu-Evenort:~$ sudo useradd -m -G dev ockner
marco@Ubuntu-Evenort:~$ sudo useradd -m -G dev wilvick
marco@Ubuntu-Evenort:~$ sudo useradd -m -G dev thiocheny
marco@Ubuntu-Evenort:~$ sudo useradd -m -G dev stanley
marco@Ubuntu-Evenort:~$ sudo useradd -m -G sec william
marco@Ubuntu-Evenort:~$ sudo useradd -m -G sec rica
marco@Ubuntu-Evenort:~$ sudo useradd -m -G sec faudia
marco@Ubuntu-Evenort:~$ sudo useradd -m -G sec wideline
marco@Ubuntu-Evenort:~$ sudo useradd -m -G sec wadsen
marco@Ubuntu-Evenort:~$ sudo useradd -m -G ops mariane
marco@Ubuntu-Evenort:~$ sudo useradd -m -G ops rico
marco@Ubuntu-Evenort:~$ sudo useradd -m -G ops frantz
marco@Ubuntu-Evenort:~$ sudo useradd -m -G ops cashna
marco@Ubuntu-Evenort:~$ sudo useradd -m -G ops jhon
marco@Ubuntu-Evenort:~$
```

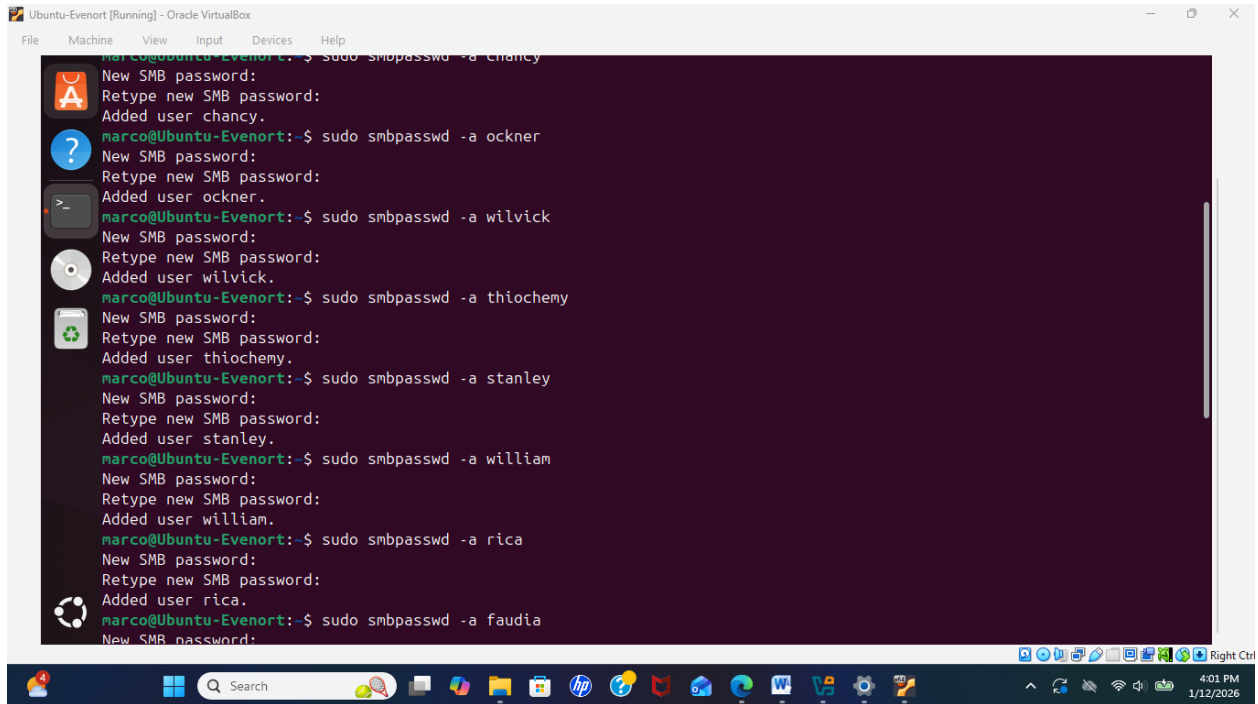
- Sur cette image, j'ai créé les trois groupes et les utilisateurs (15).

Ajout des utilisateurs à Samba:

```
sudo smbpasswd -a dev1
```

```
sudo smbpasswd -a sec1
```

```
sudo smbpasswd -a ops1
```



- Dans cette image, j'ai ajouté les utilisateurs aux groupes.

5. Création des répertoires (arborescence) de partage

```
sudo mkdir -p /srv/samba/public
```

```
sudo mkdir -p /srv/samba/dev
```

```
sudo mkdir -p /srv/samba/sec
```

```
sudo mkdir -p /srv/samba/ops
```

```
sudo mkdir -p /srv/samba/dropbox
```

The screenshot shows a terminal window titled 'Ubuntu-Evenort [Running] - Oracle VirtualBox'. The terminal output is as follows:

```
marco@Ubuntu-Evenort:~$ sudo smbpasswd -a rico
New SMB password:
Retype new SMB password:
Added user rico.
marco@Ubuntu-Evenort:~$ sudo smbpasswd -a frantz
New SMB password:
Retype new SMB password:
Added user frantz.
marco@Ubuntu-Evenort:~$ sudo smbpasswd -a cashna
New SMB password:
Retype new SMB password:
Added user cashna.
marco@Ubuntu-Evenort:~$ sudo smbpasswd -a jhon
New SMB password:
Retype new SMB password:
Added user jhon.
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
sudo: mkdir-p: command not found
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
sudo: mkdir-p /srv/samba/public: command not found
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
sudo: mkdir-p: command not found
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/dev
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/sec
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/ops
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/dropbox
marco@Ubuntu-Evenort:~$
```

- Cette image illustre les répertoires qui a été crée.

6. Partages d'équipe (DEV / SEC / OPS)

6.1 Partage Public

- Accessible à tous
- Lecture et écriture sans mot de passe

```
sudo chmod 777 /srv/samba/public
```

- **Privé dev:**

```
sudo chown -R :dev /srv/samba/dev
```

```
sudo chmod 2770 /srv/samba/dev
```

- **Privé sec:**

```
sudo chown -R :sec /srv/samba/sec
```


sudo chmod 2770 /srv/samba/sec

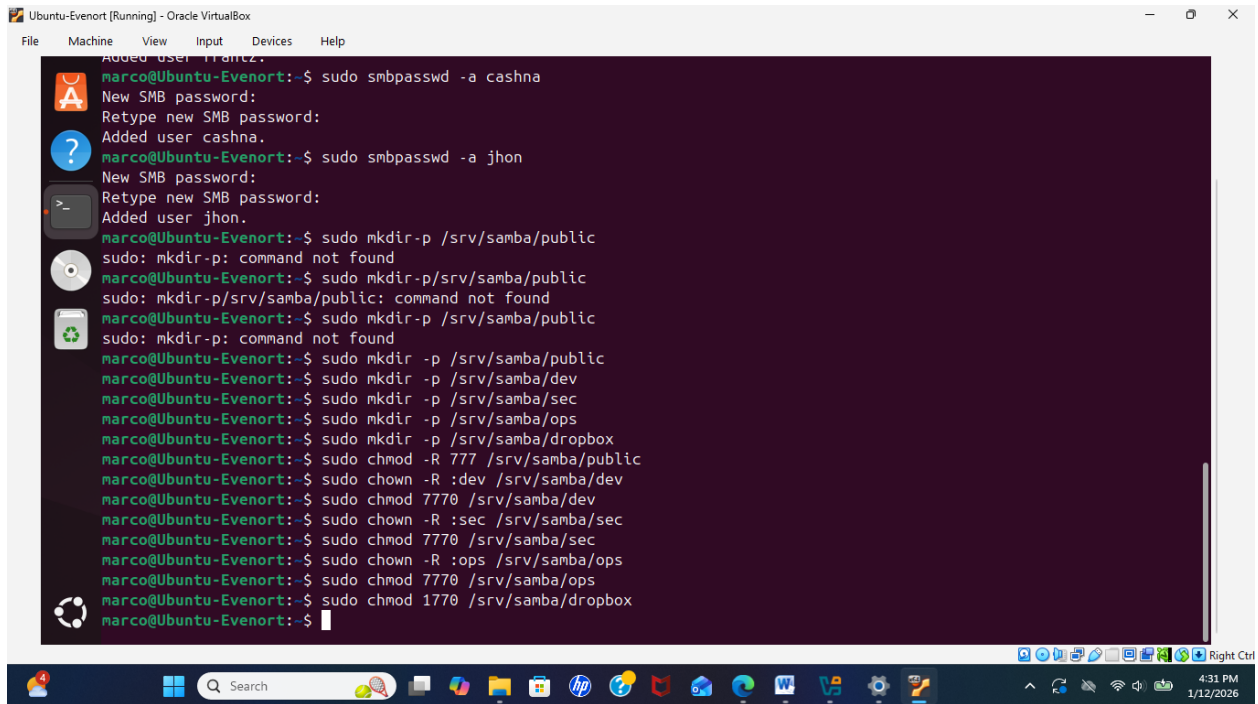
- **Privé sec:**

sudo chown -R :ops /srv/samba/ops

sudo chmod 2770 /srv/samba/ops

- **DropBox (transfert sécurisé)**

sudo chmod 1777 /srv/samba/dropbox



```
marco@Ubuntu-Evenort:~$ sudo smbpasswd -a cashna
New SMB password:
Retype new SMB password:
Added user cashna.
marco@Ubuntu-Evenort:~$ sudo smbpasswd -a jhon
New SMB password:
Retype new SMB password:
Added user jhon.
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
sudo: mkdir-p: command not found
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
sudo: mkdir-p: command not found
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
sudo: mkdir-p: command not found
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/public
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/dev
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/sec
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/ops
marco@Ubuntu-Evenort:~$ sudo mkdir -p /srv/samba/dropbox
marco@Ubuntu-Evenort:~$ sudo chmod -R 777 /srv/samba/public
marco@Ubuntu-Evenort:~$ sudo chown -R :dev /srv/samba/dev
marco@Ubuntu-Evenort:~$ sudo chmod 7770 /srv/samba/dev
marco@Ubuntu-Evenort:~$ sudo chown -R :sec /srv/samba/sec
marco@Ubuntu-Evenort:~$ sudo chmod 7770 /srv/samba/sec
marco@Ubuntu-Evenort:~$ sudo chown -R :ops /srv/samba/ops
marco@Ubuntu-Evenort:~$ sudo chmod 7770 /srv/samba/ops
marco@Ubuntu-Evenort:~$ sudo chmod 1777 /srv/samba/dropbox
marco@Ubuntu-Evenort:~$
```

7. Configuration du serveur Samba

[Public]

path = /srv/samba/public

browseable = yes

writable = yes

guest ok = yes

read only = no

[DEV]

path = /srv/samba/dev

valid users = @dev

write list = @dev

read only = no

browsable = yes

[SEC]

path = /srv/samba/sec

valid users = @sec

write list = @sec

read only = no

browsable = yes

[OPS]

path = /srv/samba/ops

valid users = @ops

write list = @ops

read only = no

browsable = yes

[DropBox]

path = /srv/samba/dropbox

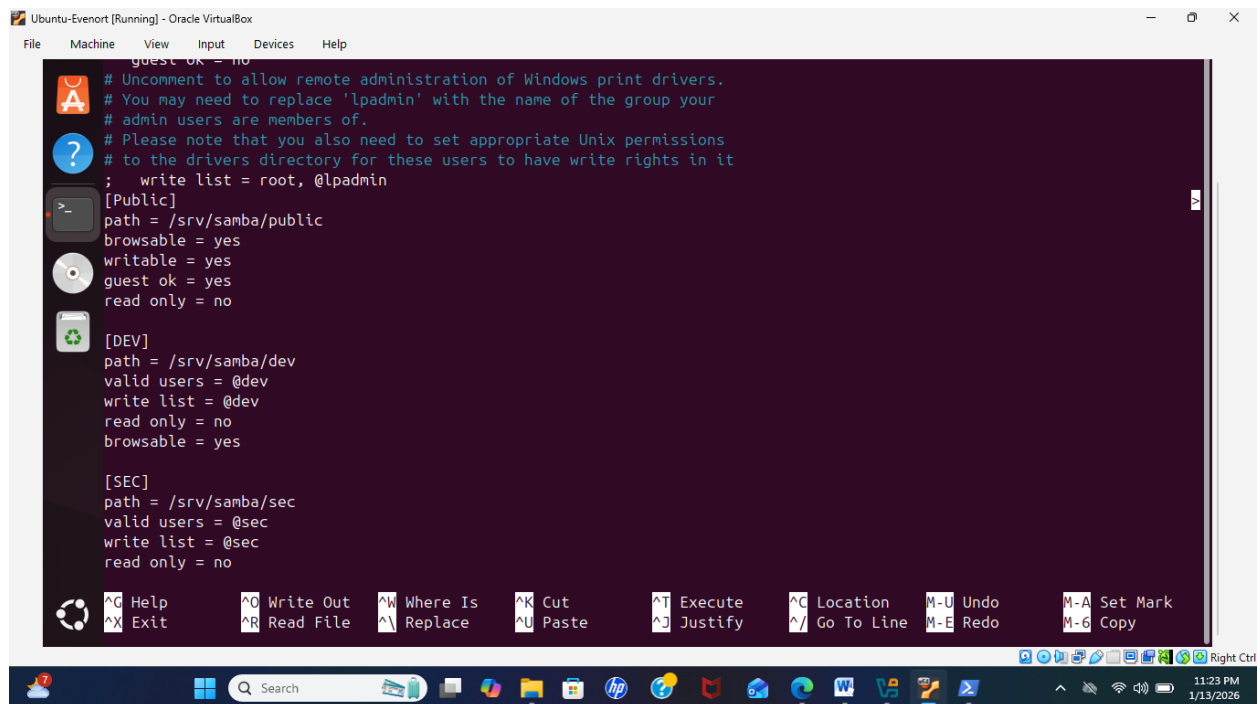
browsable = yes

writable only = yes

read only = no

guest = ok

create mask = 1777



The screenshot shows a terminal window titled "Ubuntu-Evenort [Running] - Oracle VirtualBox". The terminal displays the configuration for three Samba shares: [Public], [DEV], and [SEC]. The [Public] share is located at /srv/samba/public, is browsable, writable, and has guest access enabled. The [DEV] share is located at /srv/samba/dev, is browsable, and has valid users @dev. The [SEC] share is located at /srv/samba/sec, is browsable, and has valid users @sec. The terminal also shows some initial configuration comments and a list of keyboard shortcuts at the bottom.

```
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin

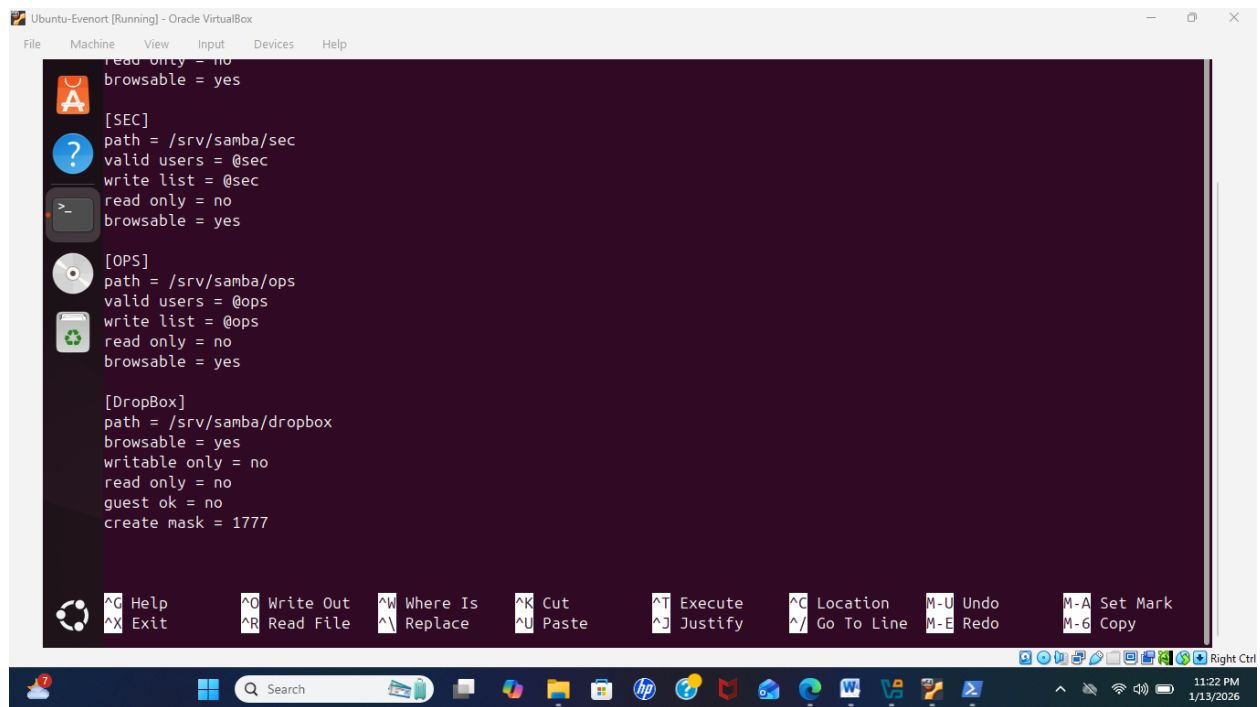
[Public]
path = /srv/samba/public
browsable = yes
writable = yes
guest ok = yes
read only = no

[DEV]
path = /srv/samba/dev
valid users = @dev
write list = @dev
read only = no
browsable = yes

[SEC]
path = /srv/samba/sec
valid users = @sec
write list = @sec
read only = no
```

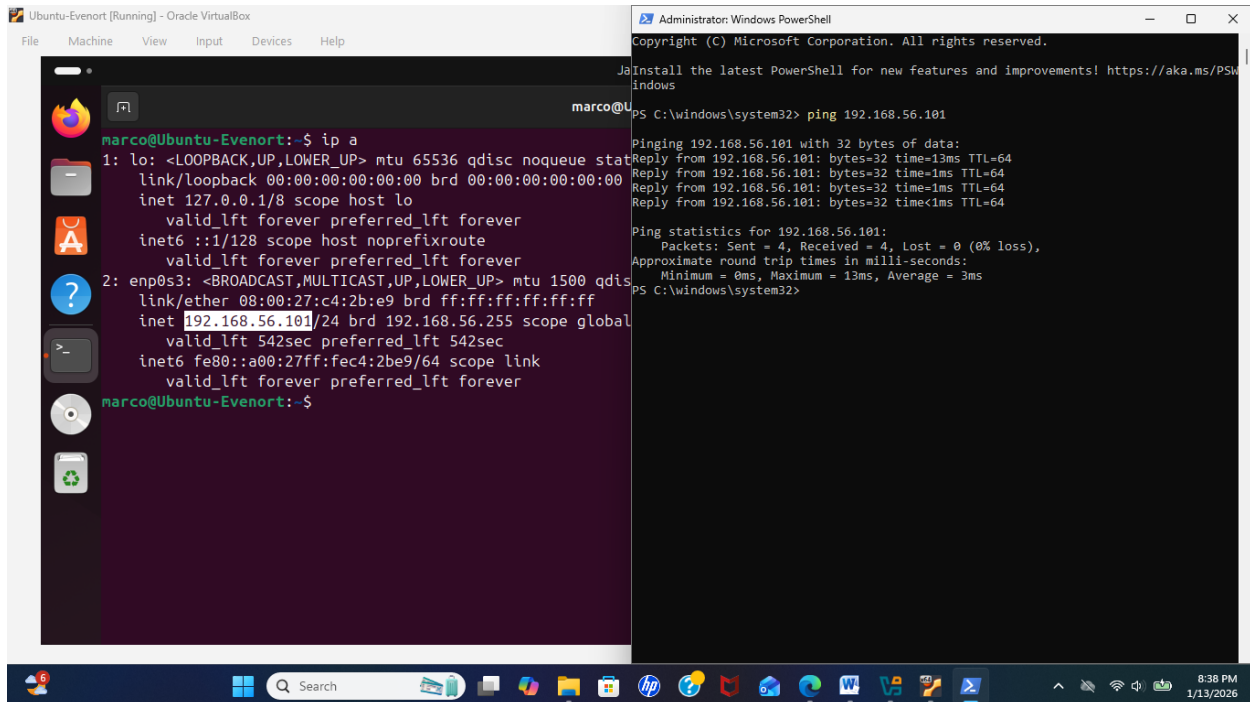
Keyboard shortcuts at the bottom of the terminal window:

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-6 Copy



- Sur cette image, j'ai configuré le serveur Samba (voir page 11 à 12).

- J'ai effectué la commande **ip a**

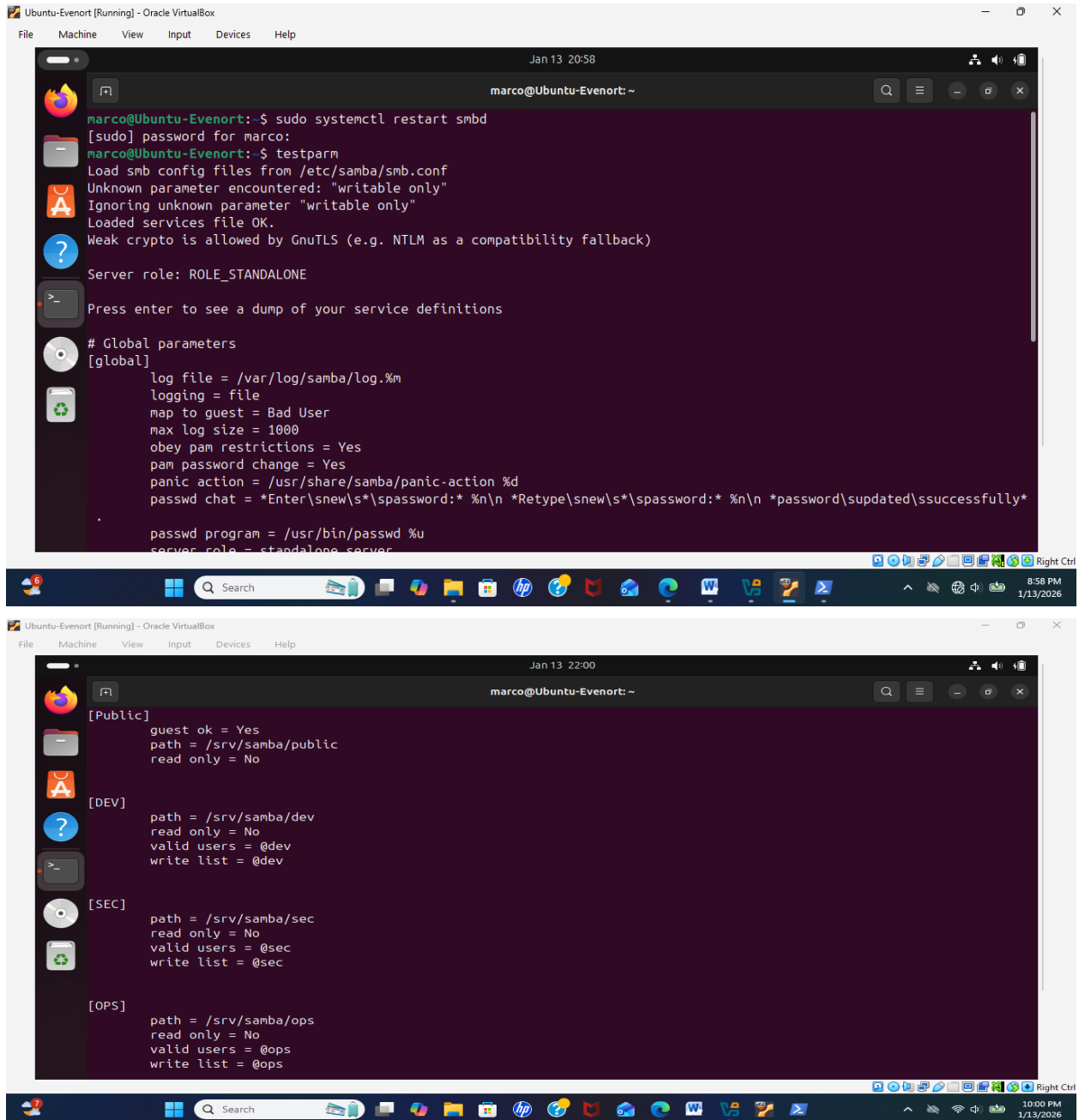


- Sur Powershell j'ai épinglé l'adresse IP de Samba, donc la connexion est établie.

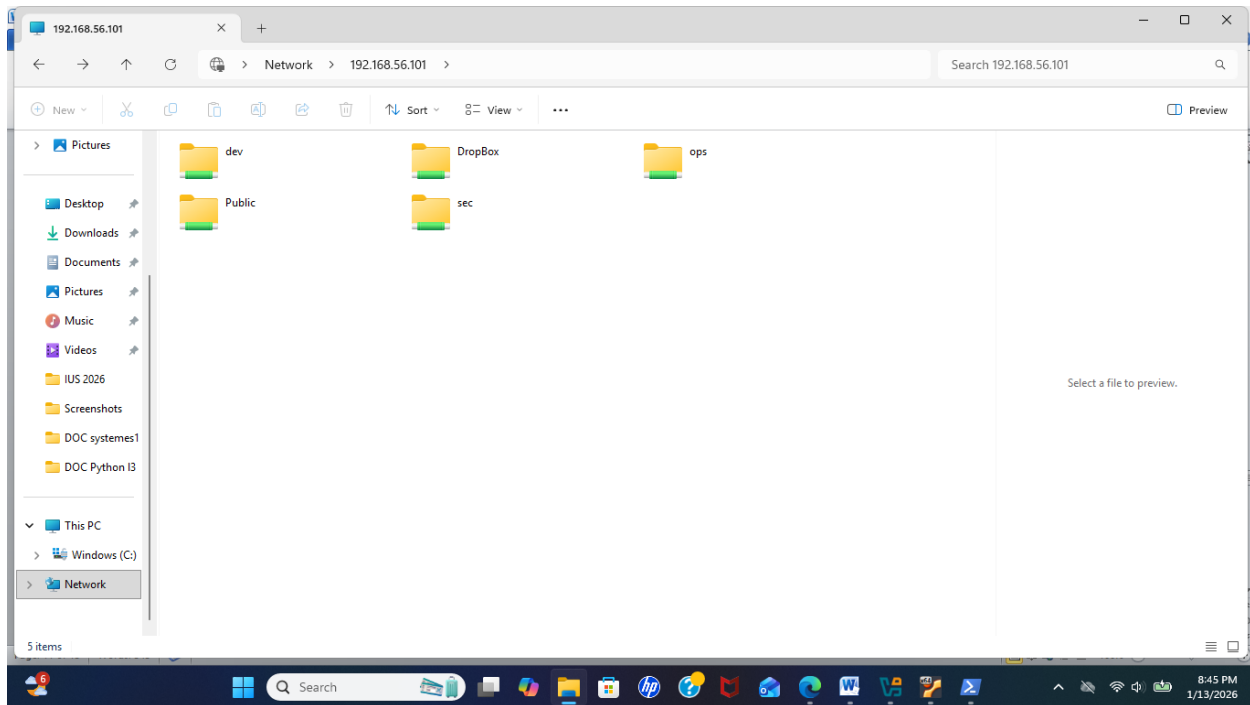
10. Redémarrer sur Samba

sudo systemctl restart smbd

testparm



- Sur ces images, j'ai redémarré samba puis j'ai fait la verification.



- Sur cette image, j'ai effectué un ping depuis mon PC afin de vérifier la connexion aux fichiers Samba.

10. Audit du serveur Samba

Commandes d'audit

testparm

smbstatus

ls -ld /srv/samba/*

ACL éventuelles:

getfacl -R /srv/samba

Erreurs Samba dans les logs

sudo tail -n 50 /var/log/samba/log.smbd

Vérification des utilisateurs Samba

sudo pdbedit -L

Vérification des connexions actives

sudo smbstatus

```
Ubuntu-Evenort [Running] - Oracle VirtualBox
File Machine View Input Devices Help

path = /srv/samba/ops
read only = No
valid users = @ops
write list = @ops

[DropBox]
create mask = 01777
path = /srv/samba/dropbox
read only = No
marco@Ubuntu-Evenort:~$ smbstatus
smbstatus only works as root!
marco@Ubuntu-Evenort:~$ sudo smbstatus

Samba version 4.19.5-Ubuntu
PID Username Group Machine Protocol Version Encryption Signi
ng
-----
3526 wilvick wilvick 192.168.56.1 (ipv4:192.168.56.1:53573) SMB3_11 - parti
al(AES-128-GMAC)

Service pid Machine Connected at Encryption Signing
-----
IPC$ 3526 192.168.56.1 Tue Jan 13 08:57:32 PM 2026 EST - AES-128-GMAC

No locked files
marco@Ubuntu-Evenort:~$
```

```
Ubuntu-Evenort [Running] - Oracle VirtualBox
File Machine View Input Devices Help

[DropBox]
create mask = 01777
path = /srv/samba/dropbox
read only = No
marco@Ubuntu-Evenort:~$ smbstatus
smbstatus only works as root!
marco@Ubuntu-Evenort:~$ sudo smbstatus

Samba version 4.19.5-Ubuntu
PID Username Group Machine Protocol Version Encryption Signi
ng
-----
3526 wilvick wilvick 192.168.56.1 (ipv4:192.168.56.1:53573) SMB3_11 - parti
al(AES-128-GMAC)

Service pid Machine Connected at Encryption Signing
-----
IPC$ 3526 192.168.56.1 Tue Jan 13 08:57:32 PM 2026 EST - AES-128-GMAC

No locked files

marco@Ubuntu-Evenort:~$ ls -ld /srv/samba/*
drwsrws--T 2 root dev 4096 Jan 12 16:08 /srv/samba/dev
drwxrwx--T 2 root root 4096 Jan 12 16:08 /srv/samba/dropbox
drwsrws--T 2 root ops 4096 Jan 12 16:08 /srv/samba/ops
drwxrwxrwx 2 root root 4096 Jan 12 16:08 /srv/samba/public
drwsrws--T 2 root sec 4096 Jan 12 16:08 /srv/samba/sec
marco@Ubuntu-Evenort:~$
```



```
Ubuntu-Evenort [Running] - Oracle VirtualBox
File Machine View Input Devices Help
drwxrwx--T 2 root root 4096 Jan 12 16:08 /srv/samba/sec
marco@Ubuntu-Evenort:~$ getfacl -R /srv/samba
getfacl: Removing leading '/' from absolute path names
# file: srv/samba
# owner: root
# group: root
user::rwx
group::r-x
other::r-x

# file: srv/samba/public
# owner: root
# group: root
user::rwx
group::rwx
other::rwx

# file: srv/samba/dropbox
# owner: root
# group: root
# flags: --t
user::rwx
group::rwx
other::---

getfacl: /srv/samba/dropbox: Permission denied
# file: srv/samba/dev
# owner: root
```

```
Ubuntu-Evenort [Running] - Oracle VirtualBox
File Machine View Input Devices Help
# group: sec
# flags: sst
user::rwx
group::rwx
other::---

getfacl: /srv/samba/sec: Permission denied
marco@Ubuntu-Evenort:~$ sudo tail -n 50 /var/log/samba/log.smbd
[sudo] password for marco:
[2026/01/12 11:45:14.117457, 0] source3/smbd/server.c:1746(main)
smbd version 4.19.5-Ubuntu started.
Copyright Andrew Tridgell and the Samba Team 1992-2023
[2026/01/12 13:08:21.962744, 0] source3/smbd/server.c:1746(main)
smbd version 4.19.5-Ubuntu started.
Copyright Andrew Tridgell and the Samba Team 1992-2023
[2026/01/12 18:03:35.202195, 0] source3/smbd/server.c:1746(main)
smbd version 4.19.5-Ubuntu started.
Copyright Andrew Tridgell and the Samba Team 1992-2023
[2026/01/12 18:03:35.206859, 0] lib/param/loadparm.c:749(lpcfg_map_parameter)
Unknown parameter encountered: "writable only"
[2026/01/12 18:03:35.206908, 0] lib/param/loadparm.c:1934(lpcfg_do_service_parameter)
Ignoring unknown parameter "writable only"
[2026/01/12 18:09:44.135050, 0] source3/smbd/server.c:1746(main)
smbd version 4.19.5-Ubuntu started.
Copyright Andrew Tridgell and the Samba Team 1992-2023
[2026/01/12 18:09:44.135797, 0] lib/param/loadparm.c:749(lpcfg_map_parameter)
Unknown parameter encountered: "writable only"
[2026/01/12 18:09:44.135816, 0] lib/param/loadparm.c:1934(lpcfg_do_service_parameter)
Ignoring unknown parameter "writable only"
```

```
marco@Ubuntu-Evenort:~$ sudo pdbedit -L
cashna:1014:
jhon:1015:
chancy:1001:
wilvick:1003:
stanley:1005:
ockner:1002:
thiochemy:1004:
william:1006:
rica:1007:
faudia:1008:
wideline:1009:
wadsen:1010:
mariane:1011:
rico:1012:
frantz:1013:
marco@Ubuntu-Evenort:~$ sudo snbstatus
sudo: snbstatus: command not found
marco@Ubuntu-Evenort:~$ sudo smbstatus

Samba version 4.19.5-Ubuntu
PID      Username   Group      Machine
-----
-----

Service  pid      Machine   Connected at      Encryption  Signing
-----
```

- Sur ces images, j'ai effectué des commandes d'audit du serveur samba (voir page 15 à 17).

Conclusion

À la fin de ce travail, le serveur Samba a été installé et configuré avec succès. Les utilisateurs ont été organisés par groupes avec des droits d'accès adaptés. Le partage public, les partages d'équipe et le dossier de transfert permettent un échange de fichiers sécurisé. Les tests effectués confirment que l'infrastructure répond aux besoins de l'entreprise TechCorp.